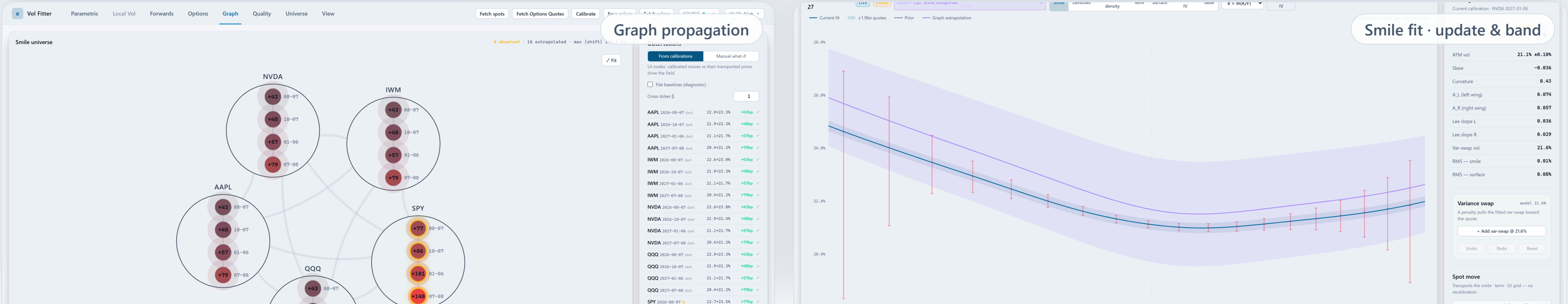


Certified volatility surfaces, with **graph propagation** to the smiles nobody quotes.

A volatility workstation that turns sparse, noisy option markets into certified, arbitrage-clean surfaces with honest uncertainty — and propagates liquid-name information across expiries and assets, with exact attribution. **The 12-slide executive cut**; a 38-slide technical deck stands behind every claim.

- 4 smile models, one calibration spine
- 1,635 tests green (1,506 backend · 129 UI)
- 22-case certification pack, client-facing
- backtested on 3 regimes + an intraday campaign



Listed option markets are **sparse, noisy, and structurally constrained** — most of the surface is never quoted.

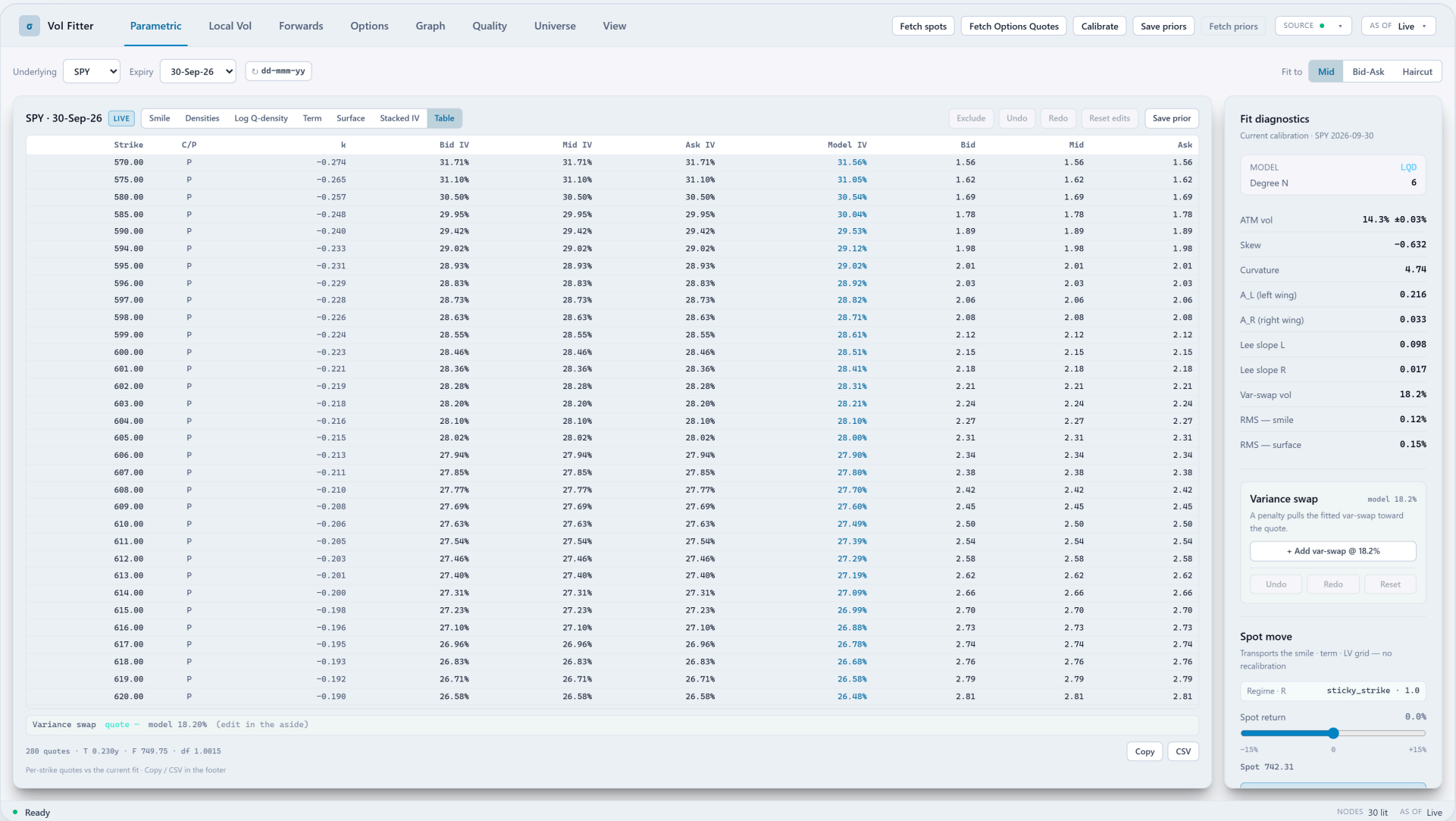
- **Wide, asymmetric spreads** — wings quote at multiples of ATM spread; mids are not fair value.
- **American exercise** — listed equity quotes embed an early-exercise premium that poisons European-model fits.
- **Structural no-arbitrage** — butterfly and calendar constraints couple every strike and expiry; a fit that ignores them produces tradable-looking nonsense.
- **Events distort the clock** — earnings compress into a day what diffusion spreads over weeks.
- **Most names are dark** — beyond the top tickers, whole expiries have no usable quotes at all.

most

of the (underlying × expiry) universe has no reliable quotes at any moment

every

downstream decision — pricing, hedging, alpha — still needs the full surface



The raw material — live SPY chain (Massive feed). Per strike: bid / mid / ask implied volatility, the model's fitted value, and the deviation. The fitter's job is to decide which of these numbers to believe, and by how much.

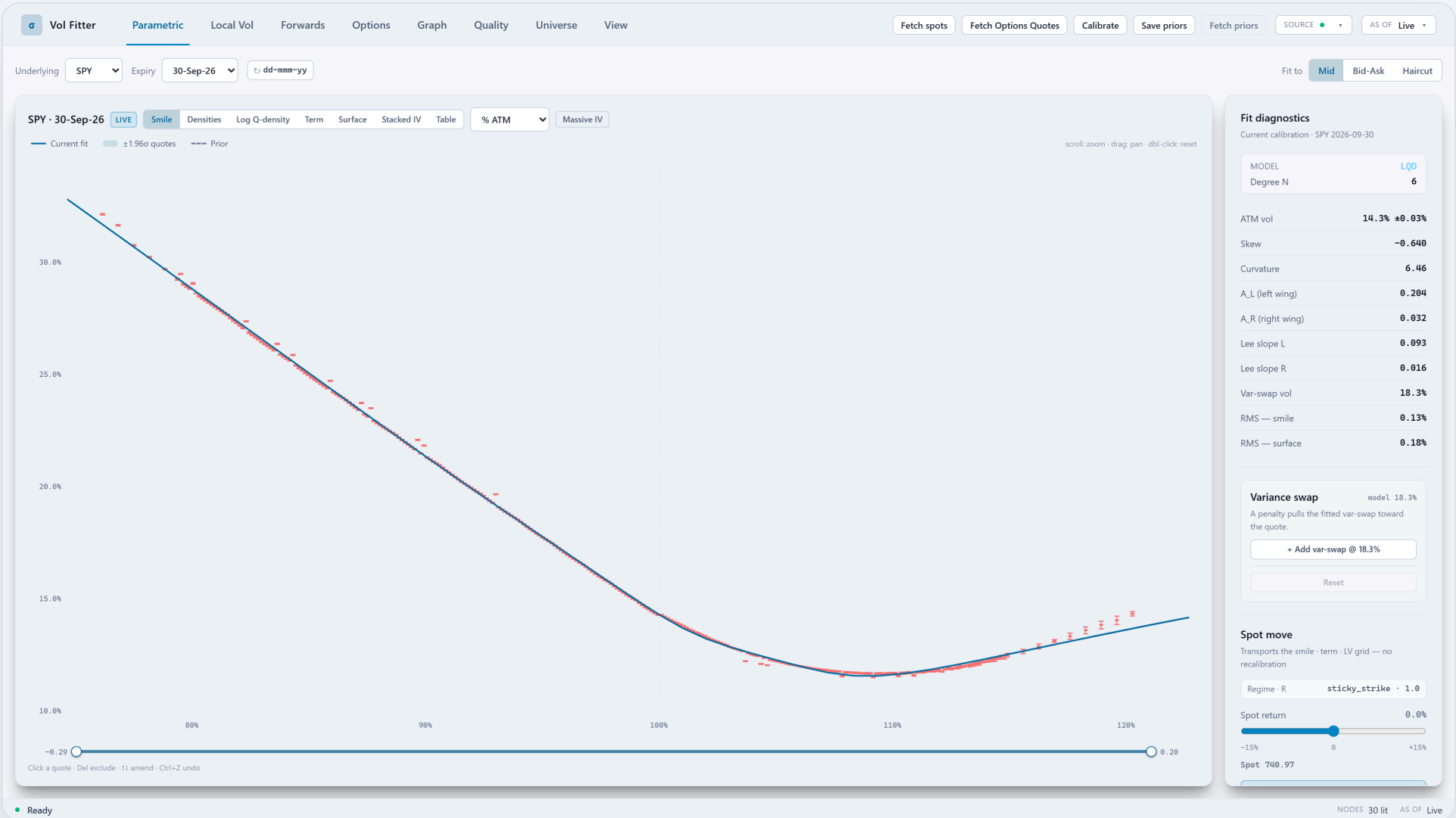
WHY IT MATTERS

Trading desks need a system that turns **partial, contradictory evidence** into a **full, arbitrage-clean surface with uncertainty**.

One quote pipeline feeds **four smile models** — all speaking the desk's own "handles": ATM vol, skew, curvature, wings.

Every quote passes one deterministic preparation pipeline (forwards, de-Americanization, liquidity weighting); every model reads and writes the same handle coordinates — which is what lets priors, filters and the graph act on any model's output without translation.

- **LQD, the house model** — parametrizes the return distribution itself, so the implied density is **positive by construction**: no butterfly arbitrage inside a slice, by design rather than by penalty.
- **SVI-JW, the market standard** — fitted in a benchmark-ratified structural chart (every iterate a valid hyperbola, wings strictly inside the Lee fence), with a per-fit **butterfly certificate** that hard-gates publish: an uncertified slice cannot become a mark.
- **Multi-Core Sigmoid, the event model** — fits the W-shaped smiles around earnings that convex models cannot reach, with hard guardrails (parameters capped by quote count, arbitrage measured on every fit).
- **Local Vol, the non-parametric surface** — a piecewise-affine local-variance grid calibrated through the pricing PDE itself; positive nodes $\Rightarrow$ positive surface $\Rightarrow$ arbitrage-free prices.



WHY IT MATTERS

Handles are the desk vocabulary — **every model, prior, filter and graph update speaks the same coordinates**, so nothing is a black box in translation.

The end goal first: SPY reprices — an unquoted NVDA smile follows, with uncertainty attached.

A controlled demo session (synthetic feed, exactly reproducible): only SPY is quoted ("lit"); QQQ, AAPL, NVDA and IWM are dark. SPY's whole surface reprices +150 bp.

1 · LIT MOVE

Every SPY quote mid is amended **+150 bp**, then SPY's four expiries are genuinely recalibrated on the shifted quotes — **real fits, not typed shifts**

2 · INNOVATION

Innovation = new fit – transported prior, per handle: **+97...+151 bp of ATM news** across SPY's expiry ladder. Only this **change** propagates — never vol levels

3 · PROPAGATION

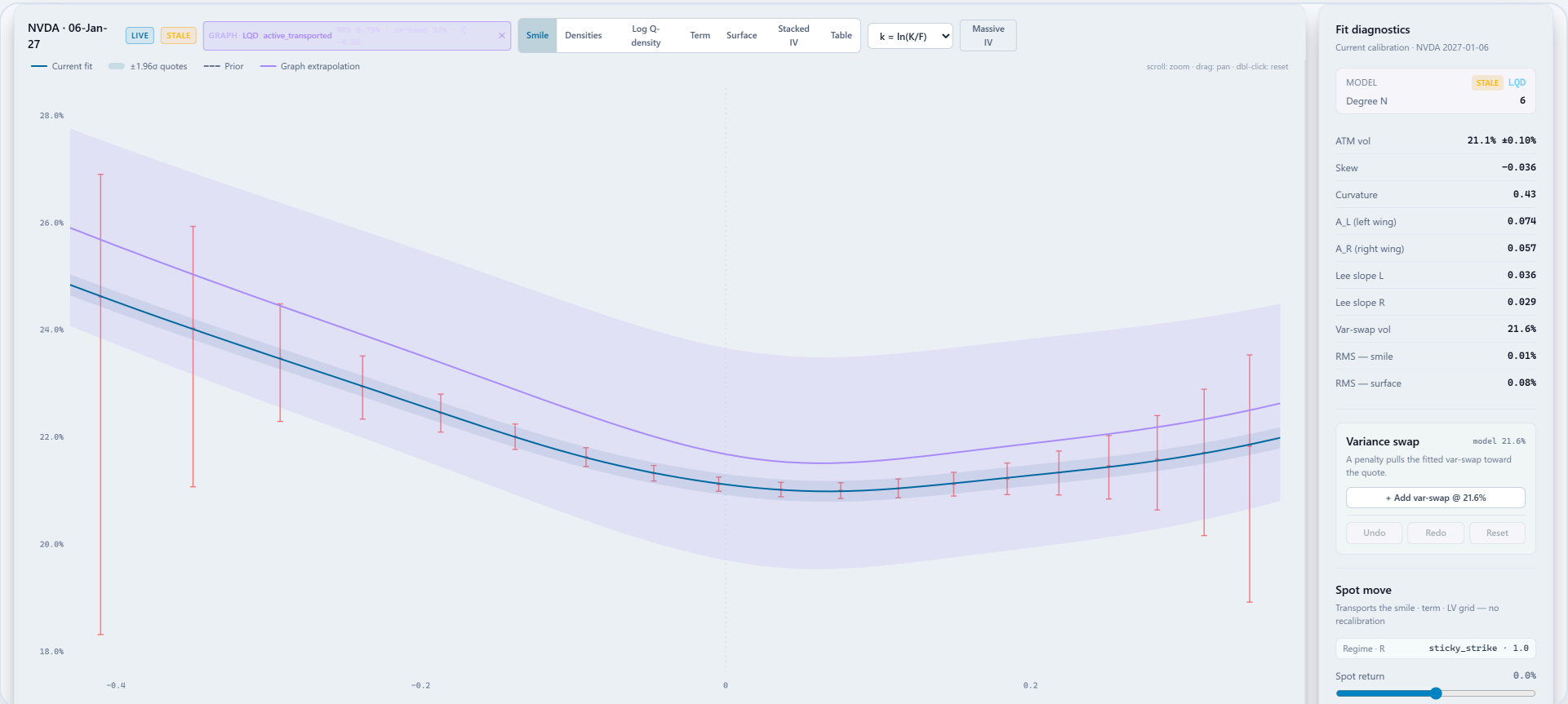
The news diffuses along calendar and cross-asset edges; each dark node blends it with its own prior by **Bayesian precision weights** — a deliberate fraction, with widened uncertainty

4 · DARK POSTERIOR

NVDA's unquoted smiles lift **+43...+79 bp of ATM vol** ($\approx \frac{1}{2}$ the SPY news), inside a **± 1.9 – 2.0 vol-pt** 95% credible band ($\pm 1.96\sigma$) they can be judged against later



The lattice after propagation. SPY's four lit nodes (amber rings) shift **+77...+140 bp**; every dark node inherits **+43...+79 bp** — per node, per expiry, largest where SPY's own move was largest



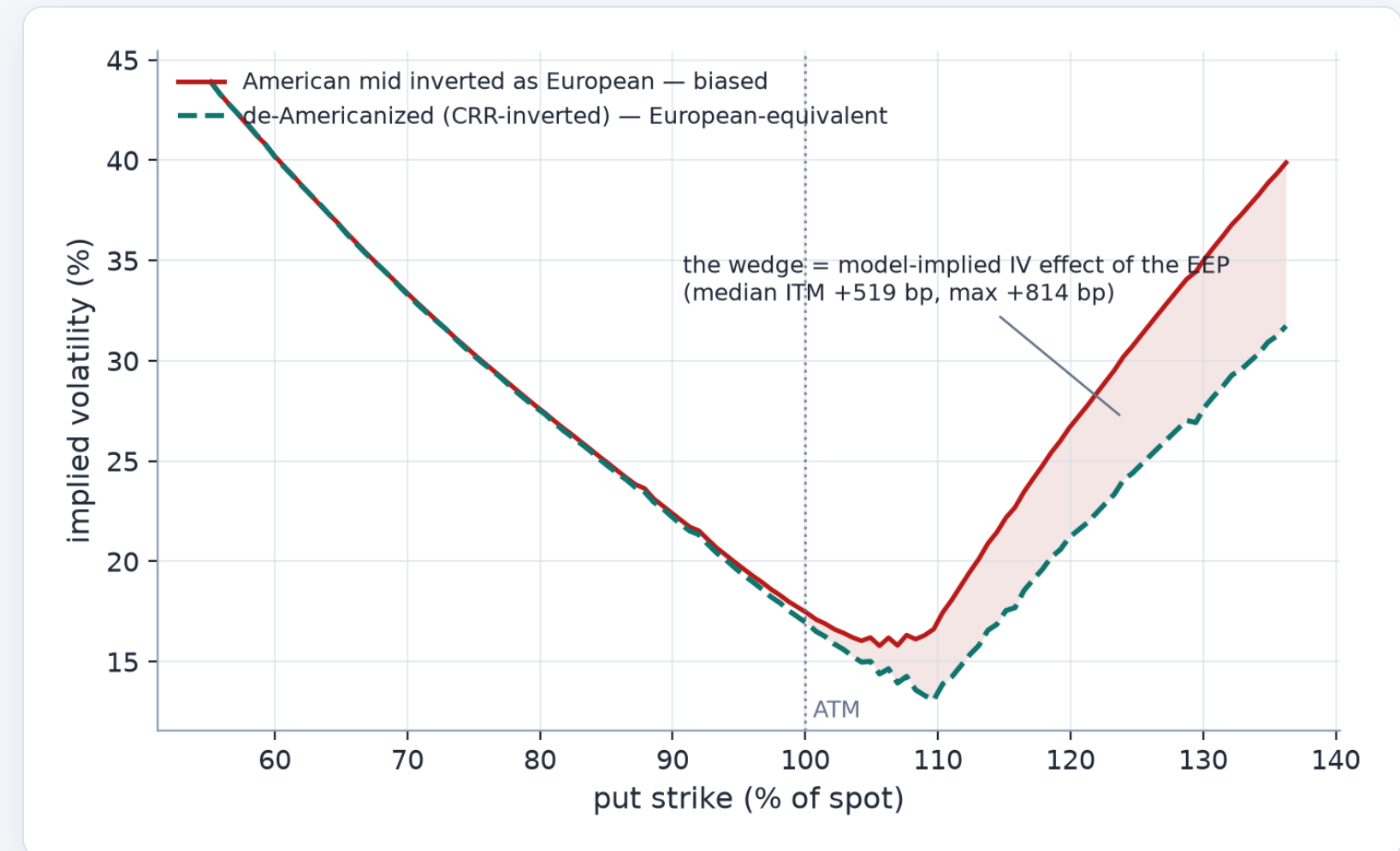
One dark smile, opened. NVDA six-month: violet — the graph posterior, **+57 bp** above the node's own last calibration (blue), with its 95% band. The badge carries the audit trail (source nodes, gain $\times$ innovation).

WHY IT MATTERS

The unquoted majority of the surface universe stays marked, moves for stated reasons, and carries stated uncertainty.

Before any model sees a quote: **early exercise removed, forwards stabilized, spreads respected.**

- **De-Americanization** — US-listed quotes embed an early-exercise premium; each is inverted through a binomial tree to its European-equivalent vol. On real SPY data (chart) the implied-vol effect on in-the-money puts runs a **median +519 bp** — a stress exhibit on the side production discards; the fitted OTM population sees a median ~4 bp, tens in the put wing. Compiled kernel: **~3 ms where NumPy takes ~350 ms.**
- **Forwards implied from the chain itself** — put-call parity as a regression; the poorly-identified discount is clamped to a physical band. A production case file: a delayed feed's stale pairs tore the ATM smile — the clamp healed it, byte-identical on clean chains.
- **Fit to the band, not the mid** — a mid inside a wide wing spread is not fair value, so staying anywhere inside the quoted spread is free and only leaving it costs. Noisy wing prints stop dragging kinks into the smile; three targets (mid / band / haircut) span the styles desks actually use.



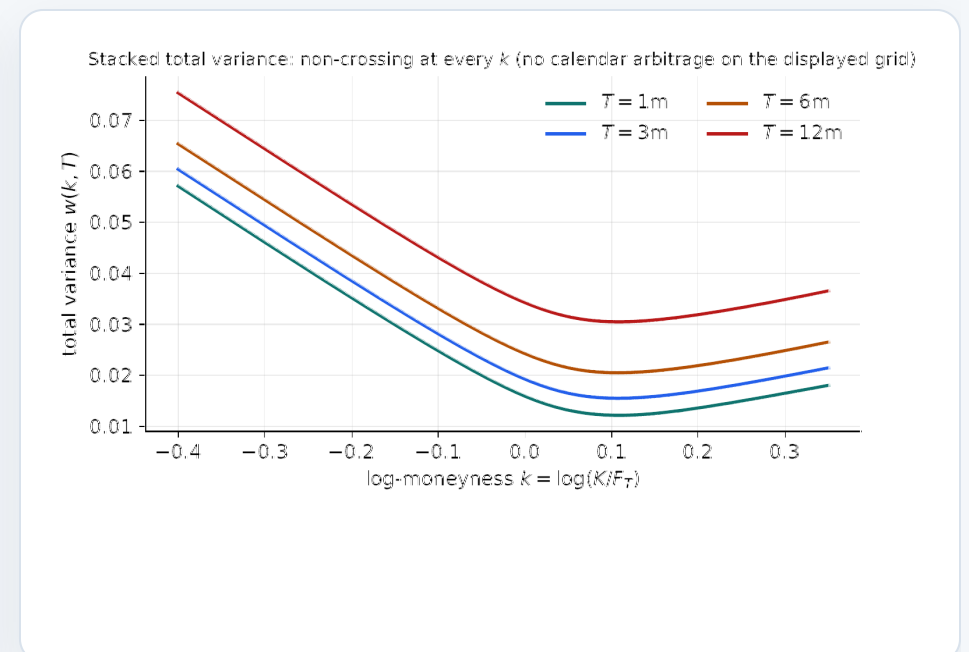
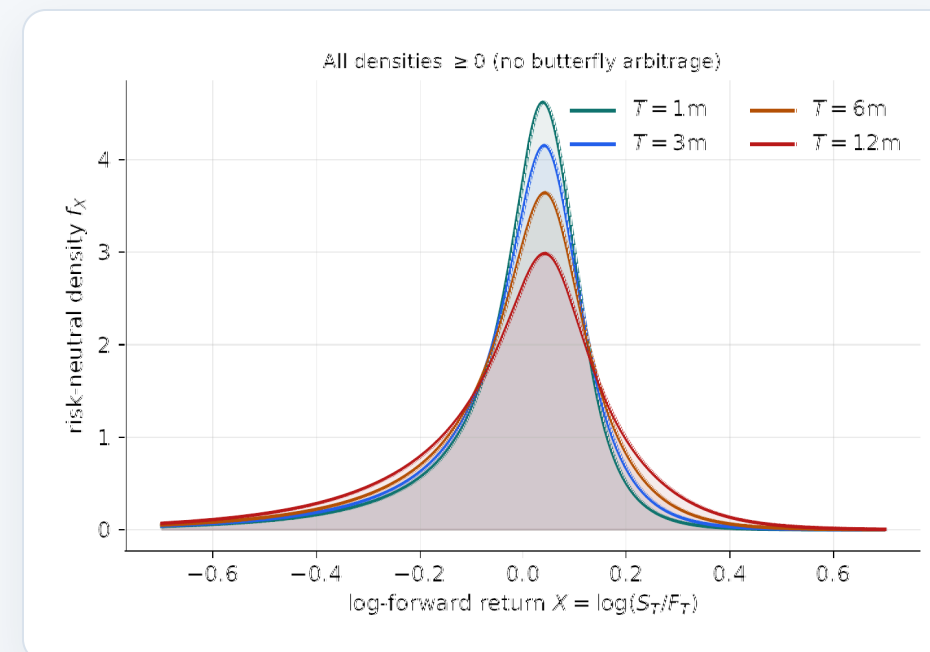
Real market data, not a toy. One live SPY expiry (Massive capture, ~6 months): every put mid inverted two ways — as if European (red) vs de-Americanized (teal). Out of the money they agree to a few bp; in the money the model-implied IV effect of the premium runs a median **+519 bp**, up to **+814 bp** — a stress exhibit on the side production discards; the fitted OTM population sees a median ~4 bp, tens in the put wing.

WHY IT MATTERS

Single names and dividend payers become fittable, every IV stands on a defensible forward, and **the fitter only pays to leave the quoted spread** — exactly how a trader thinks about it.

Arbitrage-free means **measured and enforced where data lives** — never invented where it doesn't.

- **Cleanliness is a number, then a certificate** — every fit reports its butterfly diagnostic (the Durrleman $g(k)$); SVI/MCS carry an 801-point dense-grid **certificate**, and publish is hard-blocked (HTTP 409) on an uncertified slice — a bank-committee review's sharpest demand, shipped.
- **Penalties are exactly zero on clean surfaces** — enforcement is byte-identical to OFF when the market is admissible: a clean fit is never distorted.
- **Production case files became design rules** — a calendar check sampled in unquoted strikes manufactured 151 bp of phantom penalty and flattened far-dated fits (fix: constraints confined to the traded range); the event model once invented a put wing on illiquid chains (fix: a measured, ablation-tested fence). Both are certification cases now.
- **Wings obey moment logic** — every model's tail extrapolation is a stated contract under Lee's universal cap, fenced at a **buffered 1.95**: at the bound exactly a slice can pass every screen yet price negative tail density.



Regenerated on every build. Left — all implied densities non-negative: no butterfly arbitrage, by construction. Right — stacked *total variance*, non-crossing at every strike: the actual calendar-arbitrage check, in the same coordinates as the app's Stacked IV tab (measured on independently fitted slices; enforcement is Note 10) — produced end-to-end by the production calibrator.

WHY IT MATTERS

A surface you can quote from: **no-arbitrage enforcement that repairs violations where quoted and never distorts a clean market.**

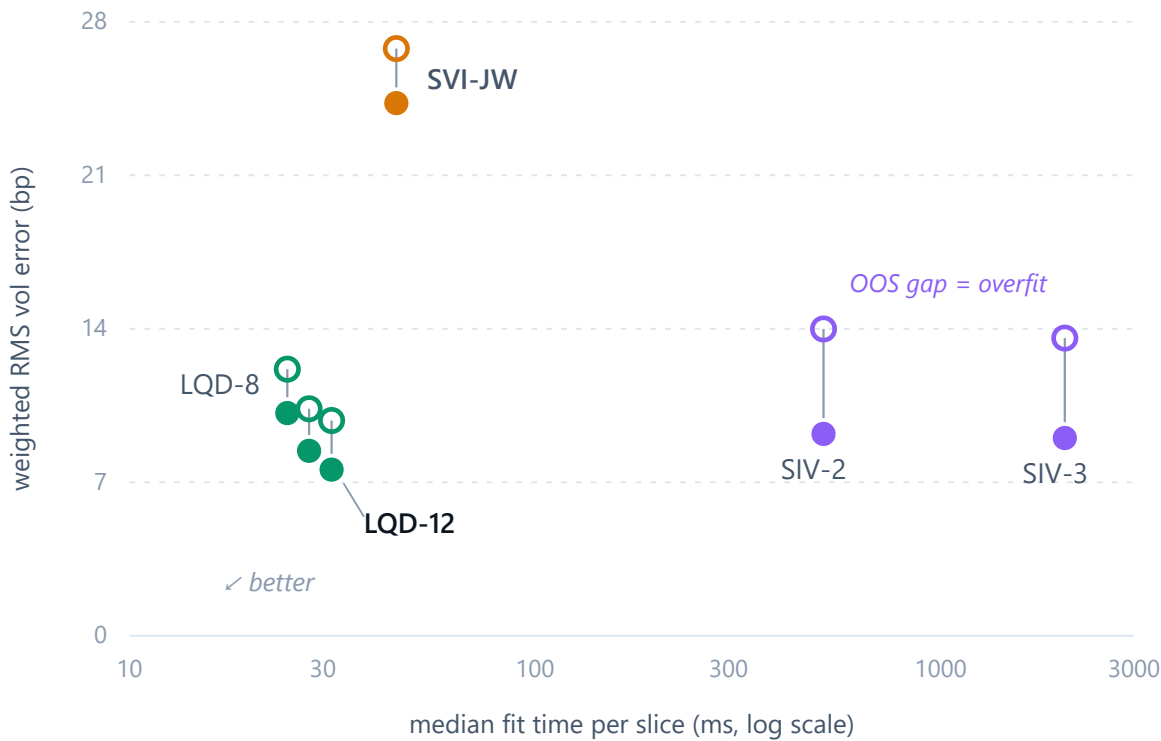
Backtested on real NBBO history: the house model at degree 8–12 fits 2–3× tighter than SVI-JW; the shipped default is ~1.6× tighter at twice SVI's speed.

Historical NBBO quotes replayed through the production calibrators over three regimes (Aug-2024 spike, high-vol Oct-2022, low-vol Jul-2023) on an 8-asset pilot universe, full expiry ladders. **Out-of-sample** = error on held-out strikes — the overfitting test.

Fit error vs fit time — spike regime (Aug 2024), 1,576 nodes

Lower-left is better · filled = in-sample RMS · hollow = out-of-sample · times pre-date SVI's analytic Jacobian (now ~10 ms)

● LQD (deg 8–12) ● SVI-JW ● Multi-Core Sigmoid (MCS) ○ out-of-sample



MODEL	IN-RMS BP	OOS BP	FIT MS	VS SVI
SVI-JW	24.3	26.8	45.5	1.00×
LQD-8	10.2	12.2	24.5	0.42×
LQD-12	7.6	9.8	31.5	0.31×
MCS-3	9.0	13.6	2023	0.37×

0.31–0.45×

LQD-12 error vs SVI, consistent across all three regimes

0

fit failures across all models and regimes

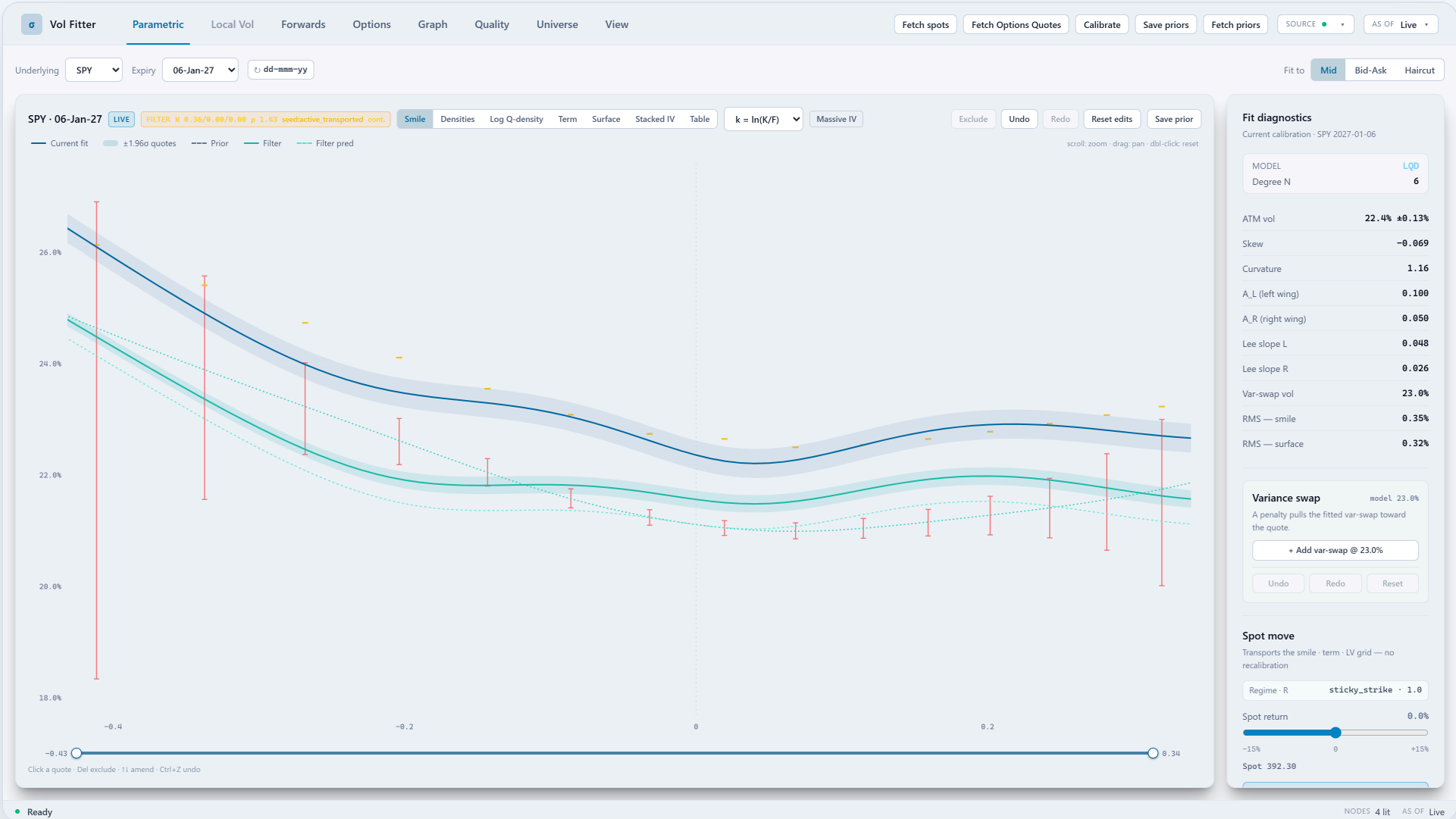
- Out-of-sample tracks in-sample for LQD — no measured overfit over the tested degrees (6–12).
- Where a model overfit (MCS's extra cores), the backtest caught it — production caps and regularizes accordingly.

WHY IT MATTERS

On replayed market history the house model fits up to 2–3× tighter than the market-standard baseline (degree 8–12; ~1.6× at the shipped default), with no measured overfit over the tested range — and the same harness reports arbitrage metrics alongside the errors.

The surface moves with spot by rule, remembers yesterday only where today is silent — and tells noise from news.

- **Spot-vol transport (SSR)** — when spot moves, every mark moves instantly by the desk's chosen stickiness rule (sticky-strike, sticky-moneyness, sticky-local-vol — one dial spans them all); recalibration then absorbs only genuine news.
- **Prior persistence** — the fit remembers yesterday's smile *shape*, *not its level*, and each pull is gated to **exactly zero** wherever today's quotes speak. Validated on history: **~32 bp** better held-out wing reconstruction, **66% win rate** over 1,116 nodes (Aug-2024 spike regime; cross-regime rerun queued) — and two candidate designs that never beat "no prior" are reported, not hidden.
- **The observation filter** — a Kalman filter on the handles separates quote noise from genuine surface motion, believing today in proportion to its measured precision. Tuned by **39,190 backtested steps**; on days when quotes contradict the chain it beats the raw fit outright (**5.6 vs 9.8 bp**) with honest uncertainty bands; on outright shock days the raw fit still wins — the flagged fix before active-by-default.



— today's raw fit — filter posterior (shaded: $\pm 1\sigma$) - - - prediction (yesterday, transported) ■ bid/ask quotes

The filter on a smile (controlled demo session). The badge is the audit: per-handle gains $K = 0.36 / 0.00 / 0.00$ — the filter believes today's ATM level about a third, and today's noisy skew and curvature not at all, so the posterior sits between prediction and fit.

WHY IT MATTERS

Marks stop chasing microstructure and sparse afternoons stop destroying morning marks — **the surface moves when the market moves, and holds when only the quotes wiggle.**

The propagation layer carries **news, never levels**, to unquoted smiles — and is validated by leave-one-out on history.

Nodes are (underlying, expiry) smiles; edges couple adjacent expiries and related names. In the default **precision-message** mode each edge is one Gaussian pairwise factor — an amplitude (the contract) and a precision (the trust, measured) — and only calibration **innovations** propagate, with **exact per-source attribution**: every dark move decomposes into named lit-node contributions. **Leave-one-out (LOO)**: pretend a lit node is dark, reconstruct it, compare to the calibration it actually had.

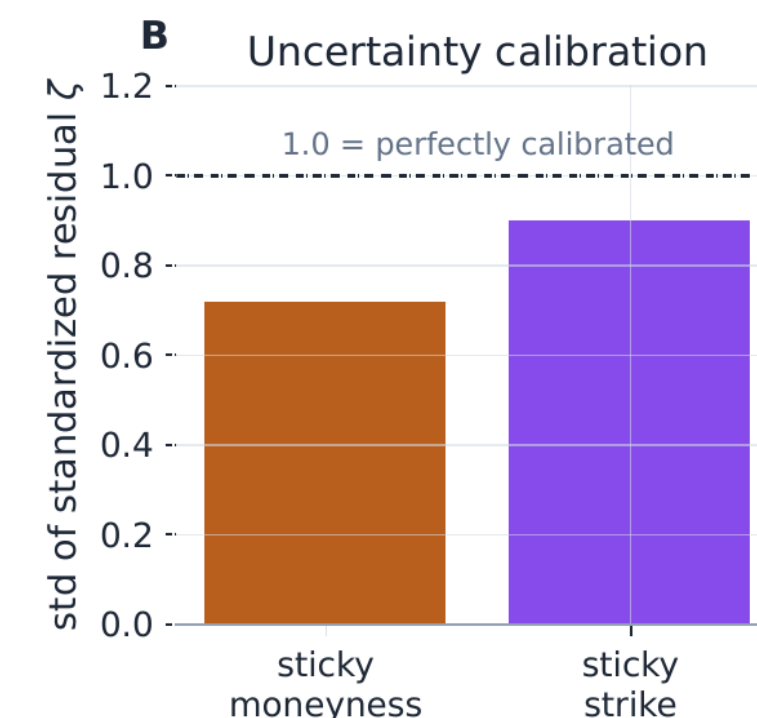
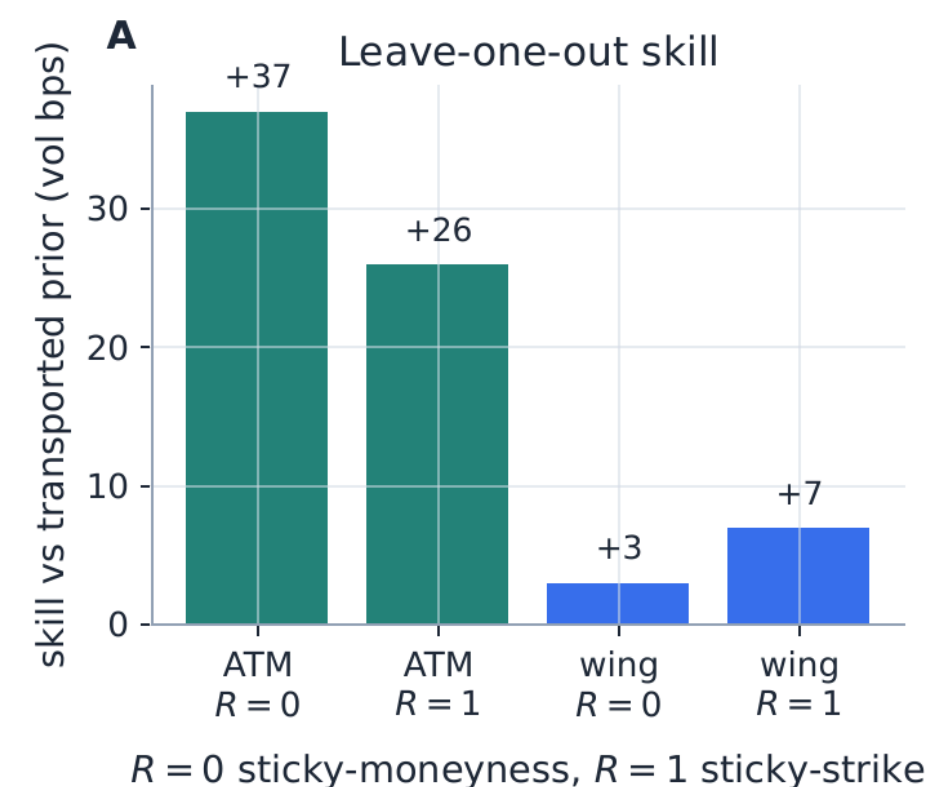
+8...+14 bp

fully-dark single names behind lit indexes/ETFs — Aug-2024 spike, at assertive reach (25 assets × 3 regimes, 47k scores)

65.8 vs 168.6 bp

intraday replay: message operator vs smooth field after a mid-session repricing — the pre-registered evidence the default flipped on

- **Decisive skill where the graph has support** — neighbour-supported indexes +10...+76 bp, ETFs +3...+7 bp across all three regimes; bands honest in *every* regime (ζ std ~ 1 after the idiosyncratic floor).
- **Honest scope** — the dark-name numbers are at assertive reach (tuned on the spike; the shipped default is deliberately timid), and in calm, idiosyncratic tape the skill is ≈ 0 , never negative. Reach is a desk dial; the one-click LOO is how a desk picks it with evidence.
- **Every edge is a number a trader can defend** — weights and per-edge β ("NVDA moves $1.3\times$ QQQ") live in a persisted policy editor with live preflight — and defaults flip only on pre-registered gates, with holds recorded as loudly as wins.

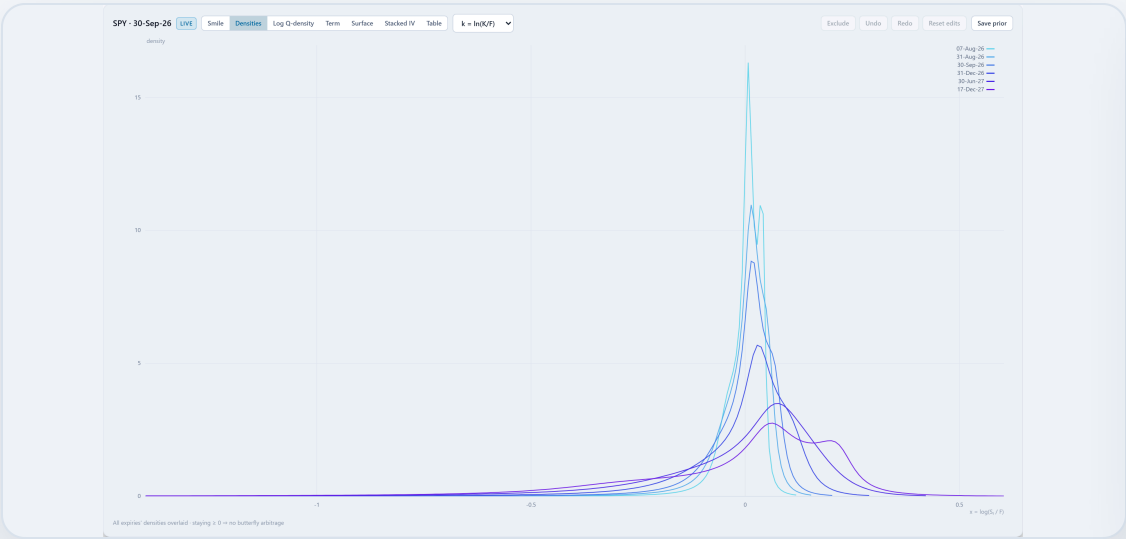


Leave-one-out, plotted. Held-out node predictions versus realized calibrations: the graph posterior tightens errors over the transported-prior baseline, and its standardized residuals sit where a correct model's should.

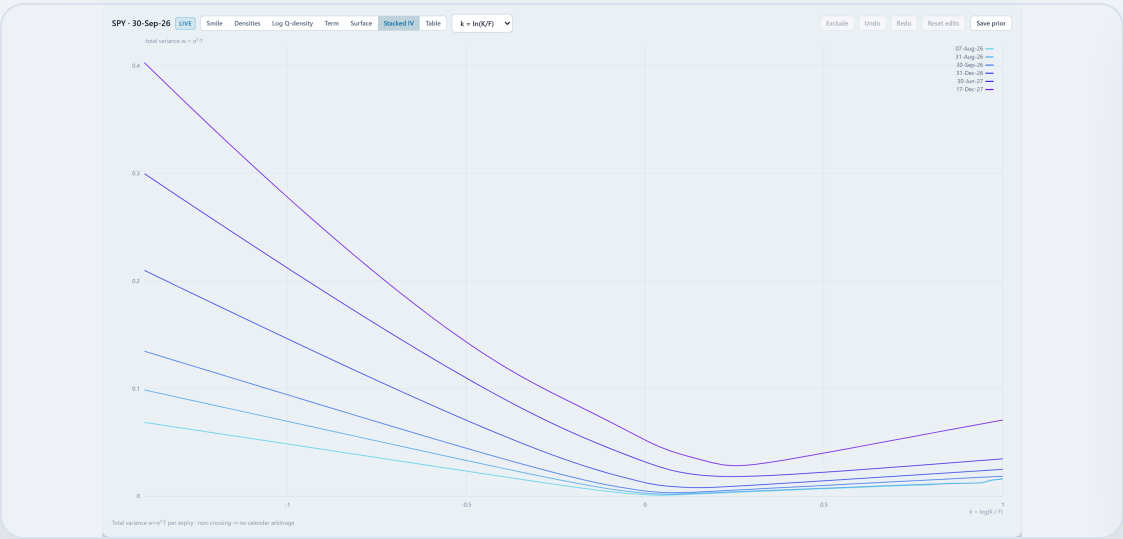
WHY IT MATTERS

The graph formalizes what traders do informally — "SPY repriced, so my dark midcap should too, a little" — with a posterior and a band instead of a hunch.

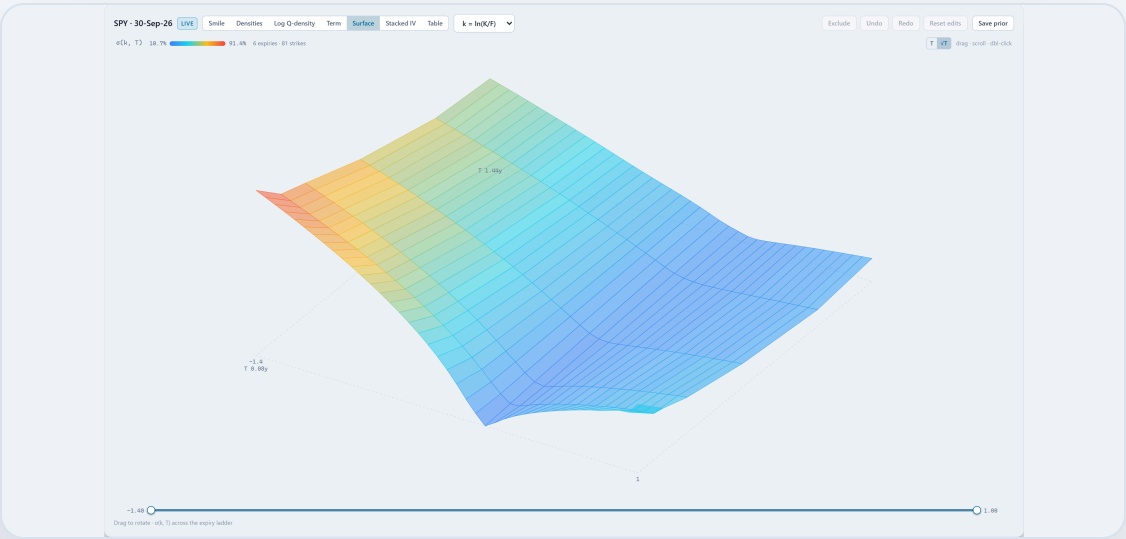
Everything above ships in a **working workstation** — eight workspaces, live against three data feeds.



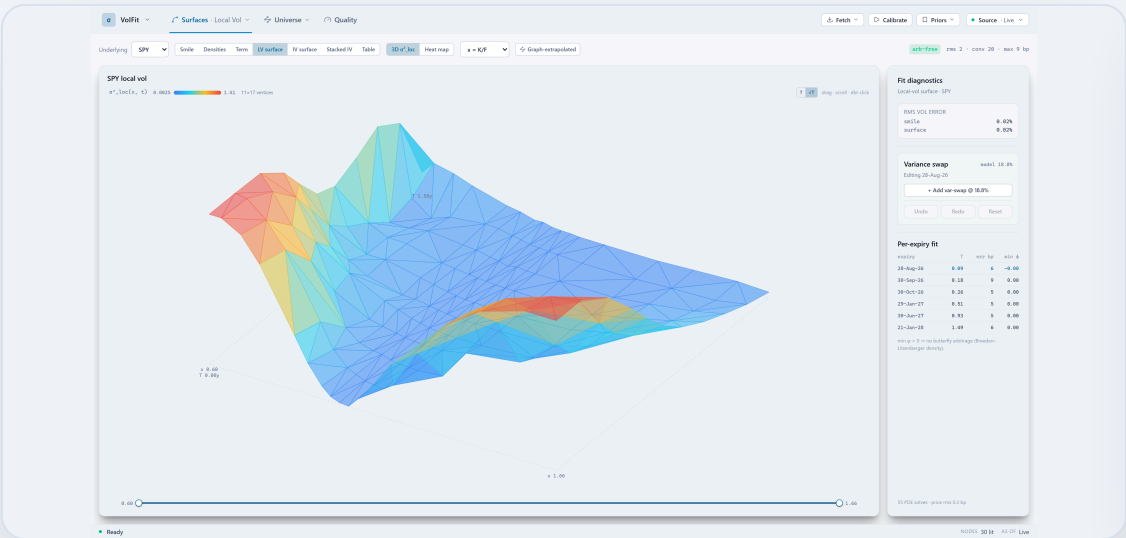
Stacked densities — every expiry ≥ 0 : no butterfly arb, at a glance



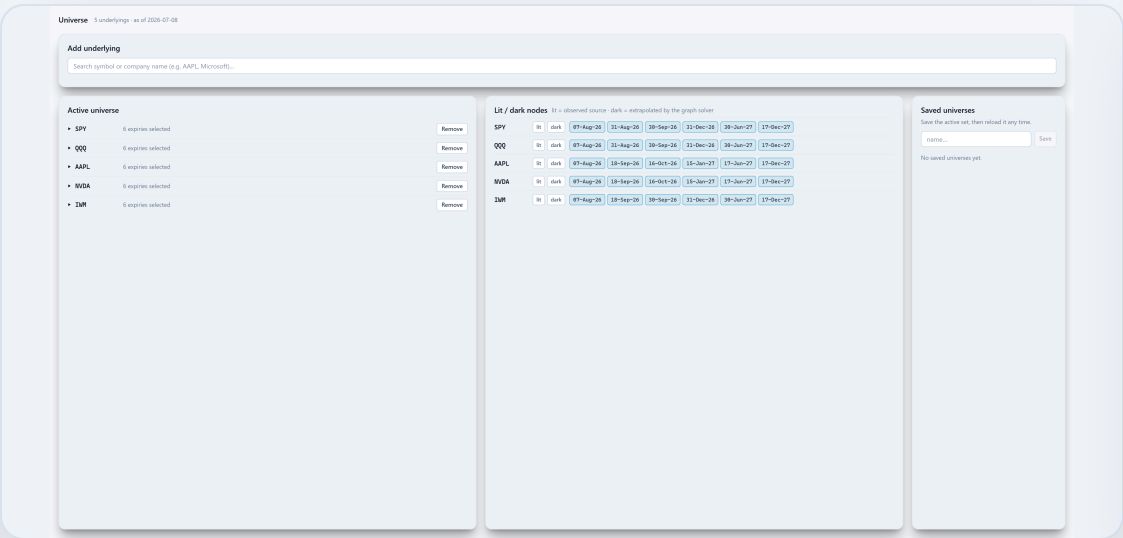
Stacked total variance — non-crossing: no calendar arb



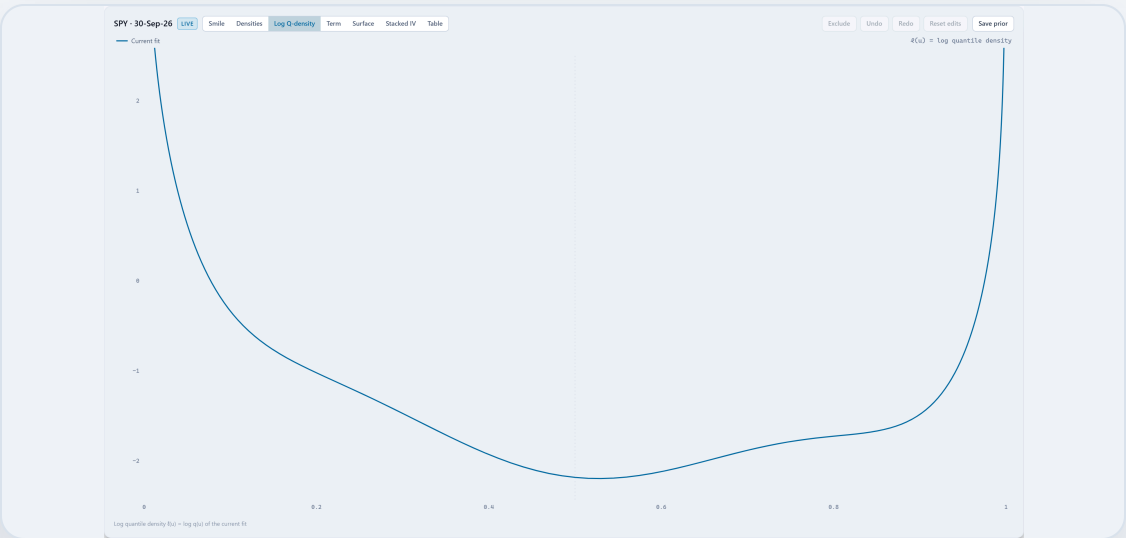
3D IV surface — rotatable $\sigma(k,T)$ mesh



3D triangulated local vol — live SPY σ^2_{loc} mesh; every triangle one exact affine element



Universe manager — catalogue, expiries, lit/dark matrix, saved universes



Log quantile density — the model's native coordinate, charted

WHY IT MATTERS

Parametric · Local Vol · Forwards · Options · Graph · Quality · Universe · View — every claim in this deck is a screen a trader can open, live against Yahoo, Bloomberg or Massive.

A daily runbook ending at a publish gate — fast enough to sit inside the quote loop.

7.6 s warm

full 30-node universe recalibration (96.6 s cold),
laptop-class hardware

~35 ms

one smile fit — the atom of every parametric
calibration (~20 ms warm 0DTE)

~3 ms · ~6×

compiled de-Americanization (vs ~350 ms NumPy)
· local-vol PDE march

byte-identical

parallel background calibration vs serial — speed
never changes a number

1 · Select & fetch

universe, expiries, lit/dark designation; spots + chains from the active source

2 · Calibrate

background per-ticker groups on a process pool (~2× warm)

3 · Inspect & adjust

smiles, densities, term, LV; amend quotes, var-swaps, events, priors, filter

4 · Propagate

graph extrapolation over the dark universe, LOO-checked

5 · Gate & publish

per-node RMS budgets, butterfly certificates, data-age staleness — hard 409 blocks with reasons;
butterfly certificates, data-age staleness — hard 409 blocks with reasons;

Vol Fitter

Parametric

Local Vol

Forwards

Options

Graph

Quality

Universe

View

Fetch spots

Fetch Options Quotes

Calibrate

Save priors

Fetch priors

SOURCE

AS OF

Live

PUBLISH READY

23/30

FITTED

30

STALE

0

NO FIT

0

ARB FLAGS

0

MEDIAN RMS

24.7 bp

WORST RMS

177.6 bp

LV SURFACES

5/5 arb-free

TICKERS

Quality report

Surfaces JSON

Surfaces CSV

mode mid · filter off · prior hybrid

Ticker	Ready	Stale	RMS bp	Arb	LV
SPY	6/6	0	14.6	0	9.1 bp
QQQ	5/6	0	34.0	0	24.7 bp
AAPL	4/6	0	77.8	0	48.2 bp
NVDA	2/6	0	74.9	0	46.0 bp
IWM	6/6	0	21.6	0	17.7 bp

NODES

exceptions only

Exceptions first

Refresh

	Node	Model	#Q	RMS bp	Max IV bp	ATH	Lee L/R	Cal viol	Status
AAPL	2026-10-16	Lqd	56	177.6	712.7	26.2%	0.14/0.05	0	● RMS 178bp > budget 50bp
NVDA	2026-08-07	Lqd	39	145.4	405.0	40.3%	0.10/0.06	0	● RMS 145bp > budget 50bp
NVDA	2027-12-17	Lqd	72	88.2	527.5	42.9%	0.59/0.19	3.6e-13	● RMS 88bp > budget 50bp
NVDA	2026-10-16	Lqd	61	73.0	241.0	42.0%	0.18/0.09	4.6e-14	● RMS 73bp > budget 50bp
NVDA	2026-09-18	Lqd	54	68.0	268.0	42.7%	0.15/0.07	0	● RMS 68bp > budget 50bp
AAPL	2026-09-18	Lqd	51	66.7	220.8	26.6%	0.12/0.04	0	● RMS 67bp > budget 50bp
QQQ	2026-08-07	Lqd	141	54.3	205.8	24.9%	0.07/0.03	0	● RMS 54bp > budget 50bp
QQQ	2027-12-17	Lqd	183	43.3	406.6	26.6%	0.42/0.10	9.7e-14	● ready
AAPL	2026-08-07	Lqd	35	40.8	132.7	28.1%	0.07/0.03	0	● ready
IWM	2027-12-17	Lqd	71	38.5	274.3	22.8%	0.42/0.08	6.7e-14	● ready
NVDA	2027-06-17	Lqd	70	37.1	135.7	42.7%	0.32/0.13	0	● ready
QQQ	2026-08-31	Lqd	173	34.3	228.0	24.5%	0.09/0.03	3.7e-15	● ready
NVDA	2027-01-15	Lqd	66	32.8	100.6	42.6%	0.20/0.10	0	● ready
AAPL	2027-01-15	Lqd	72	31.8	109.7	26.6%	0.16/0.05	3.3e-9	● ready
SPY	2026-08-07	Lqd	127	27.2	150.1	13.4%	0.05/0.01	0	● ready
AAPL	2027-12-17	Lqd	88	22.1	101.3	28.2%	0.27/0.08	0	● ready
QQQ	2027-06-30	Lqd	89	21.5	64.9	26.5%	0.13/0.03	1.4e-7	● ready
IWM	2027-06-30	Lqd	83	18.4	42.7	22.5%	0.27/0.05	0	● ready
QQQ	2026-09-30	Lqd	274	18.2	111.9	24.8%	0.11/0.03	0	● ready
SPY	2027-12-17	Lqd	201	17.3	51.9	18.9%	0.39/0.06	4.0e-14	● ready
IWM	2026-09-30	Lqd	79	17.0	37.8	20.5%	0.09/0.02	4.8e-14	● ready
AAPL	2027-06-17	Lqd	59	16.8	45.8	27.5%	0.22/0.06	0	● ready
QQQ	2026-12-31	Lqd	111	15.1	42.2	25.7%	0.16/0.04	0	● ready
IWM	2026-09-18	Lqd	34	15.0	31.9	20.6%	0.08/0.02	0	● ready
IWM	2026-12-31	Lqd	84	14.9	41.1	21.5%	0.15/0.04	0	● ready
IWM	2026-08-07	Lqd	70	13.7	38.0	20.0%	0.04/0.01	0	● ready
SPY	2026-09-30	Lqd	200	12.1	66.9	14.3%	0.10/0.02	0	● ready
SPY	2026-08-31	Lqd	200	10.4	40.0	13.6%	0.07/0.01	0	● ready
SPY	2026-12-31	Lqd	177	10.1	27.8	15.9%	0.16/0.02	0	● ready
SPY	2027-06-30	Lqd	179	8.6	22.2	17.9%	0.22/0.03	6.6e-14	● ready

WHY IT MATTERS

The runbook a vol desk actually operates — every performance number states its hardware and cache state, held by perf rails in the suite; a 22-case certification pack (22/22 green, client-facing report) freezes every historical bug as a named, re-runnable case.

VOL-FITTER

A professional-grade fitter, plus the **graph propagation layer** sparse volatility markets are missing.

Four arbitrage-aware smile models · trading-grade quote preparation · Bayesian memory and Kalman filtering in one coordinate system · a propagation layer with exact attribution, adjudicated by pre-registered backtests, that carries information to the smiles nobody quotes. 1,600+ tests green · a 22-case client-facing certification pack · 16 technical notes + 19 lecture editions · backtested on three regimes and an intraday campaign — every number traces to an equation, a module, and a test. **The modeling stack is built and validated; what remains is productization, not invention.**